

BSc (Hons) in Software Engineering (4 year)

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SE 1307 - Programming 02

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Coursework (TEST DOCUMENT)

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Batch : 2nd Bat

Assignment Cover Sheet

Qualification		Module Number and Title
BSc (Hons) in Software Engineering (4 year)		SE 1307 - Programming 02
Student Name & No.		Assessor
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Hand over date		Submission Date
Assessment type	Duration/Length of Assessment Type	Weighting of Assessment
Coursework	Software Submission	60%

Learner declaration
I, <u>L.G.H.N.U.Senarathna.KD/ICBT/BSCSE/02/09</u> <name of the student and registration number>, certify that the work submitted for this assignment is my own and research sources are fully acknowledged.

Marks Awarded			
First assessor			
IV marks			
Agreed grade			
Signature of the assessor		Date	

FEEDBACK FORM
INTERNATIONAL COLLEGE OF BUSINESS & TECHNOLOGY

Module : SE 1307 - Programming 02
Student : L.G.H.N.U.Senarathna , KD/ICBT/BSCSE/02/09
Assessor : Mrs. Umesha Silva
Assignment : System for “Greenleaf Bakery”

Assessor feedback

Marks Awarded

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Green Leaf Bakery System

SOFTWARE TESTING DOCUMENT

1. Introduction

Beginning with how programs behave under scrutiny, software testing checks whether an application meets its defined needs. This paper aims to lay out a full approach for assessing the Green Leaf Bakery System, built with Java and based on file operations.

This document covers test plans along with test cases, integrates UML diagrams while applying White Box Testing methods to support system consistency and performance. Quality emerges through structured validation steps combined with visual modeling approaches included here. Reliability forms gradually via methodical checks rather than assumptions built into early stages. Techniques are chosen not at random but based on clarity they bring during inspection phases.

2. System Overview

The Green Leaf Bakery System is a Java-based application developed to manage bakery operations efficiently. The system allows users to:

- Register and log in.
- Place customer orders.
- Generate invoices.
- Manage bakery products.
- Maintain customer information.
- Manage staff activities.
- Store and retrieve data using text files.

The system aims to improve efficiency, reduce paperwork, and provide accurate order management.

3. Objectives of Testing

The objectives of testing are:

- Verify that all functions work correctly.
- Identify defects and errors.
- Ensure the system meets user requirements.
- Improve software quality.
- Ensure system reliability and performance.

4. Scope of Testing

The testing process covers the following modules:

1. Login Module
2. Registration Module
3. Customer Management Module
4. Order Management Module
5. Invoice Management Module
6. Product Management Module
7. File Handling Module
8. Manager Functions
9. Baker Functions

5. Testing Methodology

The following testing methods are used:

Black Box Testing

- Functional Testing
- Boundary Value Analysis
- Equivalence Partitioning

White Box Testing

- Statement Coverage
- Decision Coverage
- Path Coverage
- Basis Path Testing
- Cyclomatic Complexity

6. Functional Requirements

- **FR01**

The system shall allow users to register.

- **FR02**

The system shall allow users to log in.

- **FR03**

The system shall allow customers to place orders.

- **FR04**

The system shall generate invoices.

- **FR05**

The system shall store order details.

- **FR06**

The system shall allow managers to manage products.

- **FR07**

The system shall allow bakers to view orders.

7. Non-Functional Requirements

- **Performance**

The system should respond within 3 seconds.

- **Reliability**

The system should operate without failure.

- **Security**

Only authenticated users can access the system.

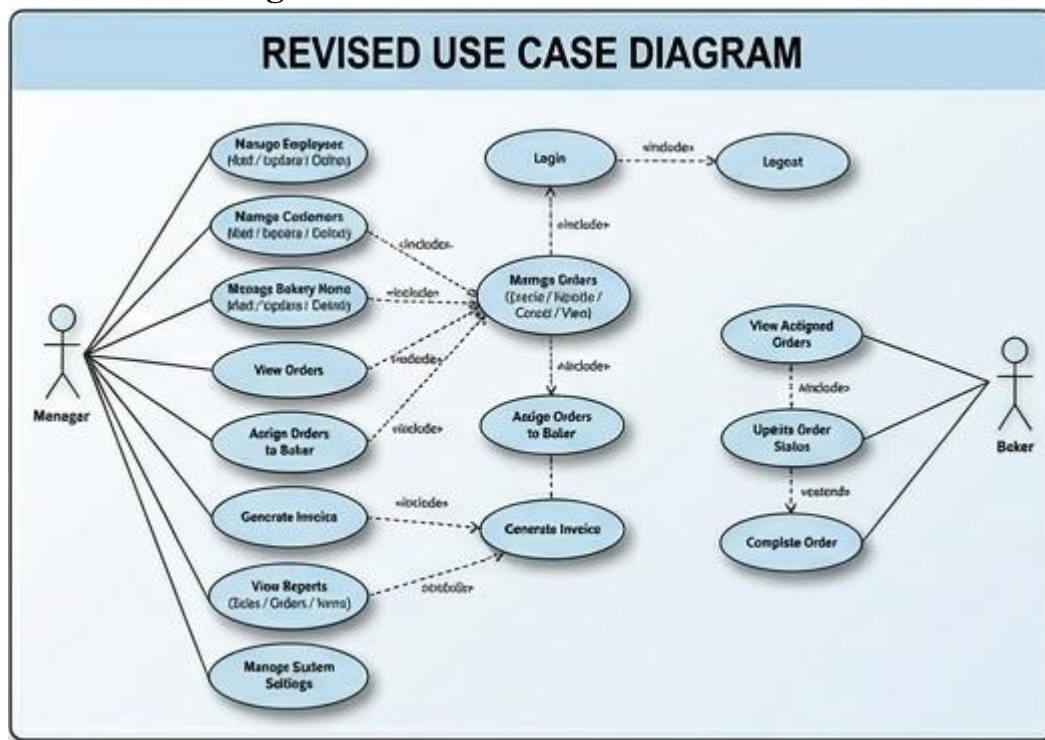
- **Usability**

The system should have a user-friendly interface.

- **Maintainability**

The system should be easy to modify and update.

8. Use Case Diagram



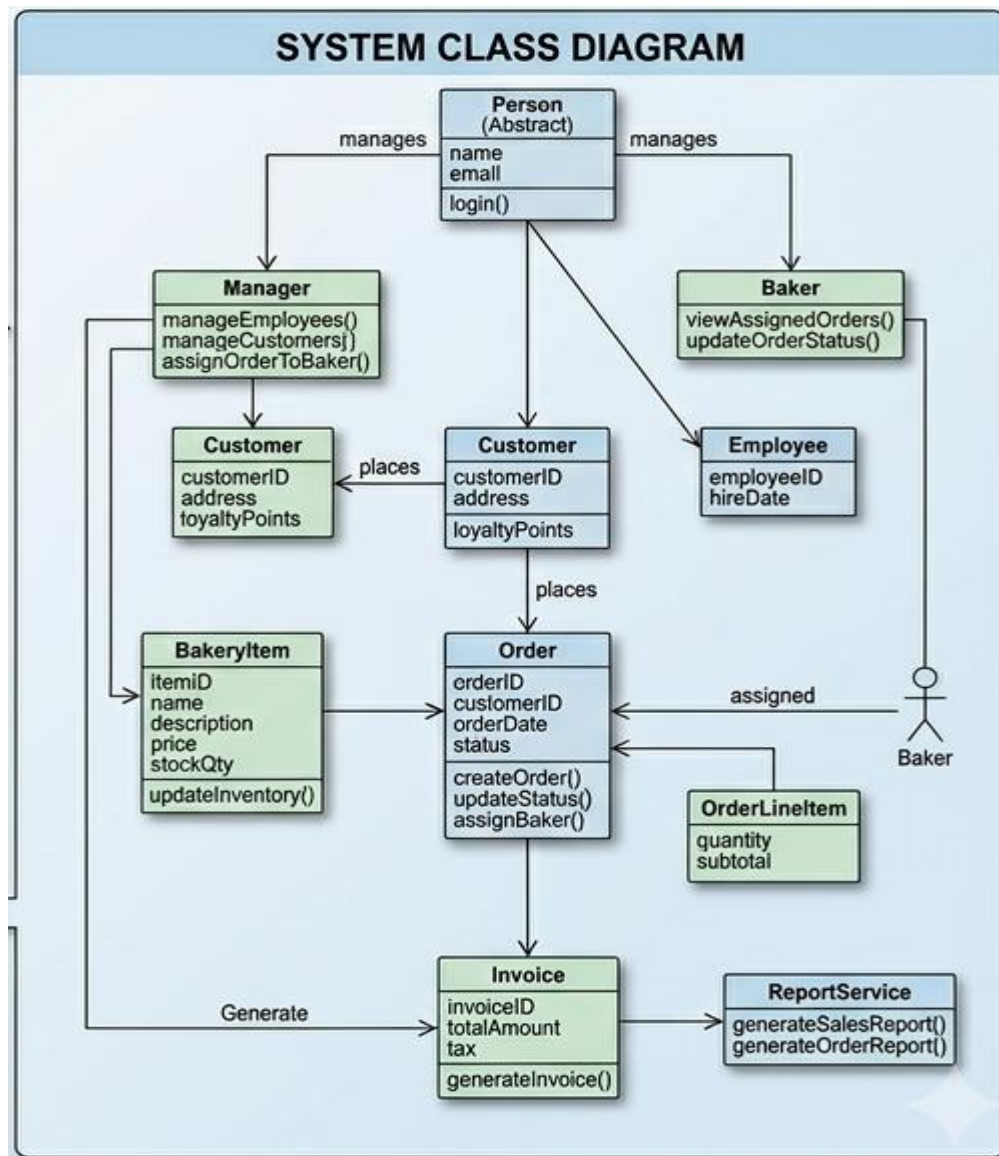
The use case diagram illustrates the interactions between

- Manager
- Baker

Main use cases include:

- Login
- Register
- Place Order
- Generate Invoice
- Manage Products
- View Reports

9. Class Diagram

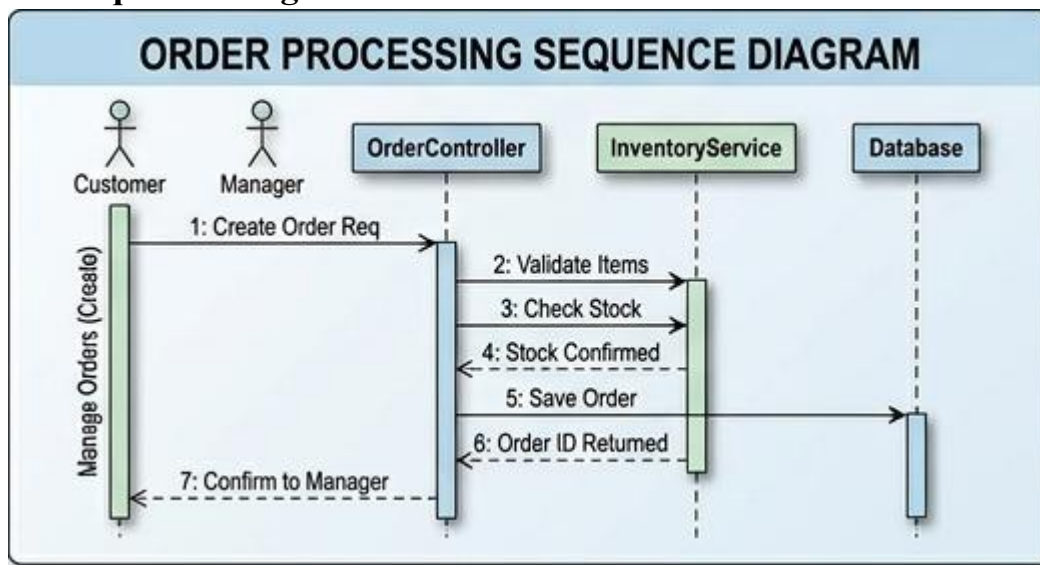


Major classes:

- User
- Customer
- Order
- Invoice
- Product
- Manager
- Baker
- File Handler

Relationships between classes are shown in the class diagram.

10. Sequence Diagram



The sequence diagram explains the order processing flow:

1. Customer places order.
2. System validates data.
3. Order is saved.
4. Invoice is generated.
5. Confirmation is displayed.

11. Test Plan

Test Item	Description
Software	Green Leaf Bakery System
Testing Type	Functional and White Box Testing
Environment	Windows 10, Java JDK, NetBeans
Tester	Student
Testing Period	-----

12. Test Cases

- **Test Case 01 – Login**

Test Case ID = TC-01

Module = Login

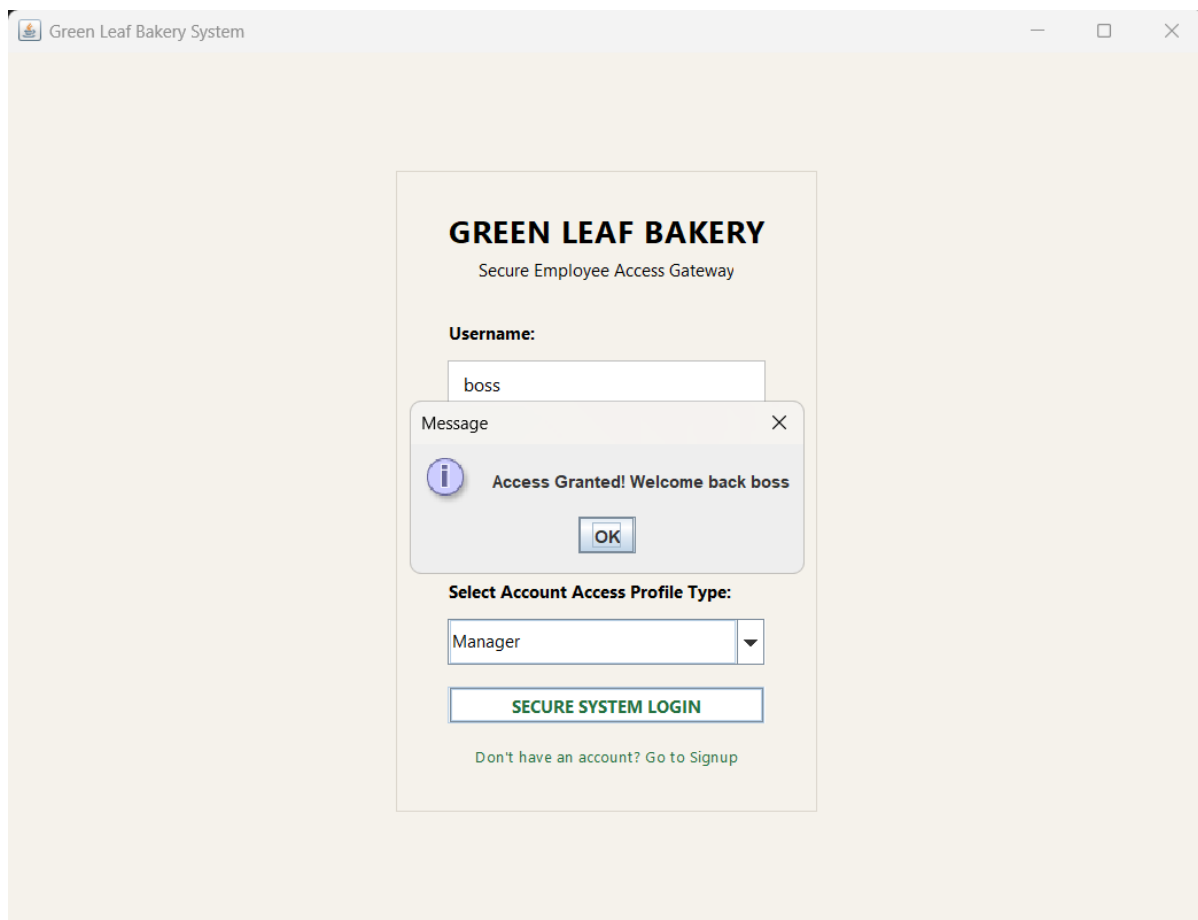
Objective = Verify user login

Input = Valid username and password

Expected Result= Login successful

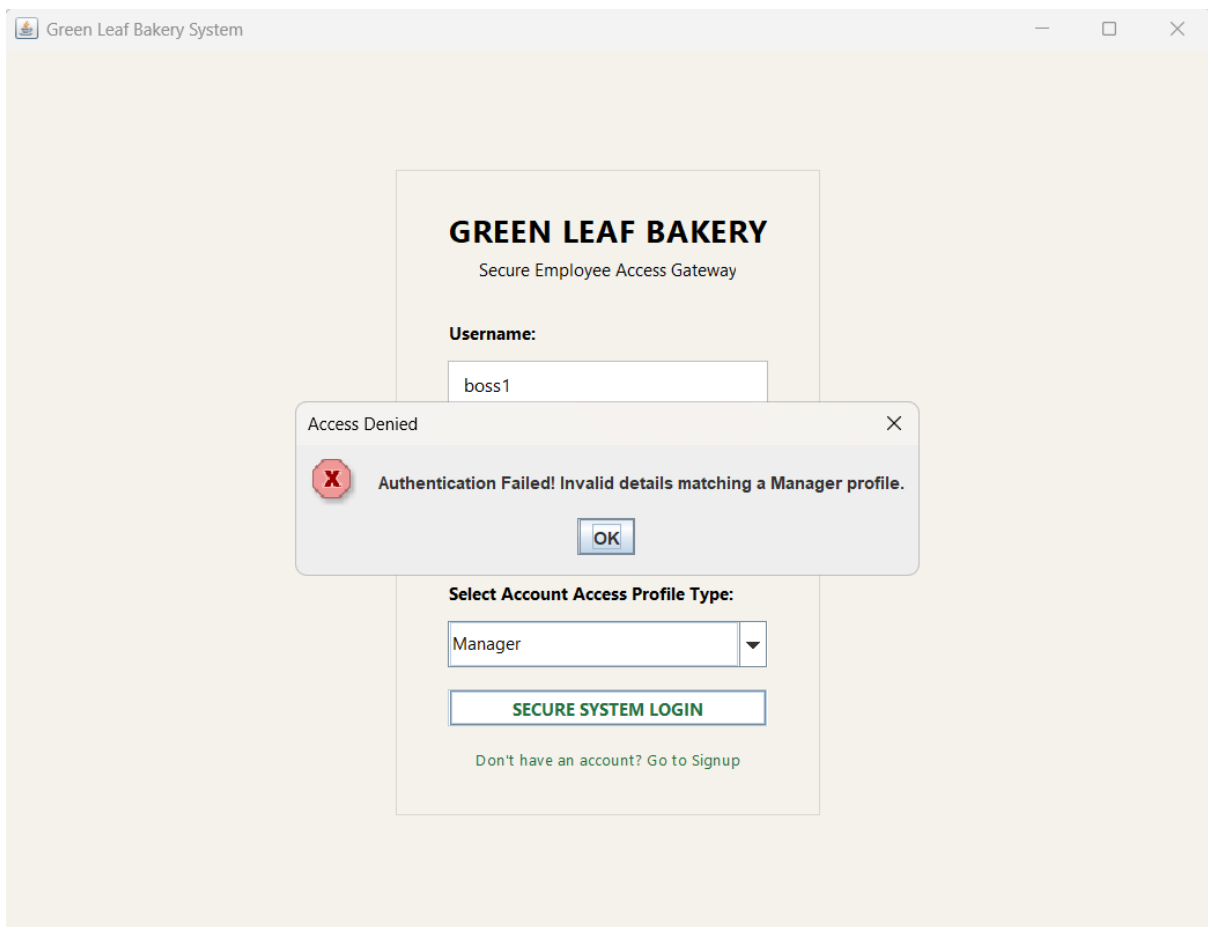
Actual Result = Login successful

Status = Pass



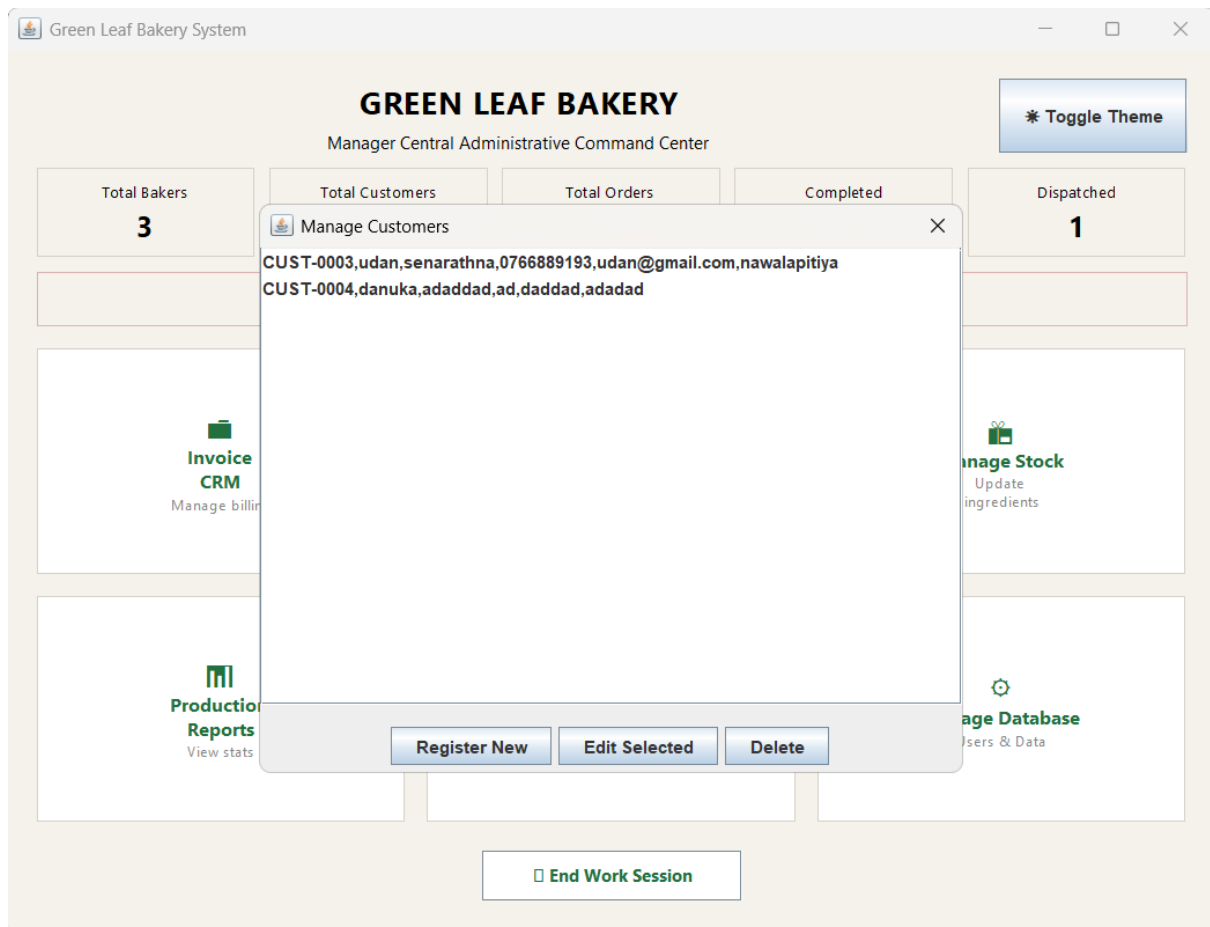
- **Test Case 02 – Invalid Login**

Test Case ID = TC-02
Module = Login
Objective = Verify invalid login
Input = Wrong username/password
Expected Result= Error message
Actual Result = Error displayed
Status = Pass



- **Test Case 03 – Customer Registration**

Test Case ID = TC-03
Module = Registration
Objective = Customer registration
Input = Customer details
Expected Result= Customer registered
Actual Result = Customer registered
Status = Pass



- **Test Case 04 – Place Order**

Test Case ID = TC-04
Module = Order
Objective = Order registration
Input = Order details
Expected Result= Order registered
Actual Result = Order registered
Status = Pass

The screenshot shows a web application window titled "Green Leaf Bakery System". The main content area is titled "GREEN LEAF BAKERY" and displays a "Dispatch Time: 2026-06-28 20:27:04". Below this, the text "Manager Central Order Production Input Desk" is visible. The form contains several fields for order entry: "Order ID" (ORD-0002), "Customer" (CUST-0003 - udan), "Details" (Email: 0766889193, Phone:), "Item Name" (cake), "Quantity" (2), "Notes" (add less suger), "Payment Status" (Pending), "Specialization" (Cakes), and "Assign Baker" (BAR-0001). A green button labeled "DISPATCH NEW PRODUCTION ORDER" is at the bottom, along with a link to "Return to Dashboard".

GREEN LEAF BAKERY

Dispatch Time: 2026-06-28 20:27:04
Manager Central Order Production Input Desk

Order ID: ORD-0002

Customer: CUST-0003 - udan

Details: Email: 0766889193
Phone:

Item Name: cake

Quantity: 2

Notes: add less suger

Payment Status: Pending

Specialization: Cakes

Assign Baker: BAR-0001

DISPATCH NEW PRODUCTION ORDER

[Return to Dashboard](#)

- **Test Case 05 – Generate Invoice**

Test Case ID = TC-05
Module = Invoice
Objective = Generate Invoice
Input = Invoice details
Expected Result= Invoice registered
Actual Result = Invoice registered
Status = Pass

The screenshot displays the 'Green Leaf Bakery System' interface. On the left, the 'Generate Order Invoices' section contains a form with the following fields: 'Inv ID' (text input with 'INV-0002'), 'Order ID' (dropdown menu with 'ORD-0001'), 'Customer' (text input with 'danuka'), 'Amount' (text input with '1500'), and 'Status' (dropdown menu with 'PENDING'). Below these fields is a green button labeled 'Issue Bakery Invoice'. To the right of this form is the 'System Records' section, which includes a 'View' filter with buttons for 'All' and 'Dispatched', a search input field, and a 'Filter' button. Below the 'Generate Order Invoices' form is a 'Quick Modifications' section with an 'ID' text input and an 'Update Status' button. A modal dialog box titled 'Message' is centered on the screen, displaying an information icon, the text 'Invoice Generated!', and an 'OK' button. At the bottom right of the interface is a button labeled 'Back to Dashboard'.

- **Test Case 06 – Save Data to File**

Test Case ID = TC-06
 Module = Save Data
 Objective = Save Data to File
 Input = Data that need to be saved
 Expected Result= Data saved into files
 Actual Result = Data saved into files
 Status = Pass

Name	Date modified	Type	Size
src	6/10/2026 2:43 PM	File folder	
target	6/10/2026 12:59 PM	File folder	
baker	6/26/2026 10:47 PM	TXT File	1 KB
customers	6/27/2026 12:50 AM	TXT File	1 KB
instructions	6/26/2026 11:17 PM	TXT File	1 KB
invoices	6/28/2026 8:29 PM	TXT File	1 KB
logins	6/28/2026 8:18 PM	TXT File	1 KB
manager	6/26/2026 10:47 PM	TXT File	1 KB
nbactions	6/13/2026 10:32 AM	Microsoft Edge HT...	3 KB
orders	6/26/2026 11:17 PM	TXT File	1 KB
pom	6/5/2026 9:07 PM	Microsoft Edge HT...	1 KB
preparation_reports	6/26/2026 11:18 PM	TXT File	1 KB
production_capacity	6/26/2026 11:19 PM	TXT File	1 KB
stock	6/26/2026 11:17 PM	TXT File	1 KB

13. White Box Testing

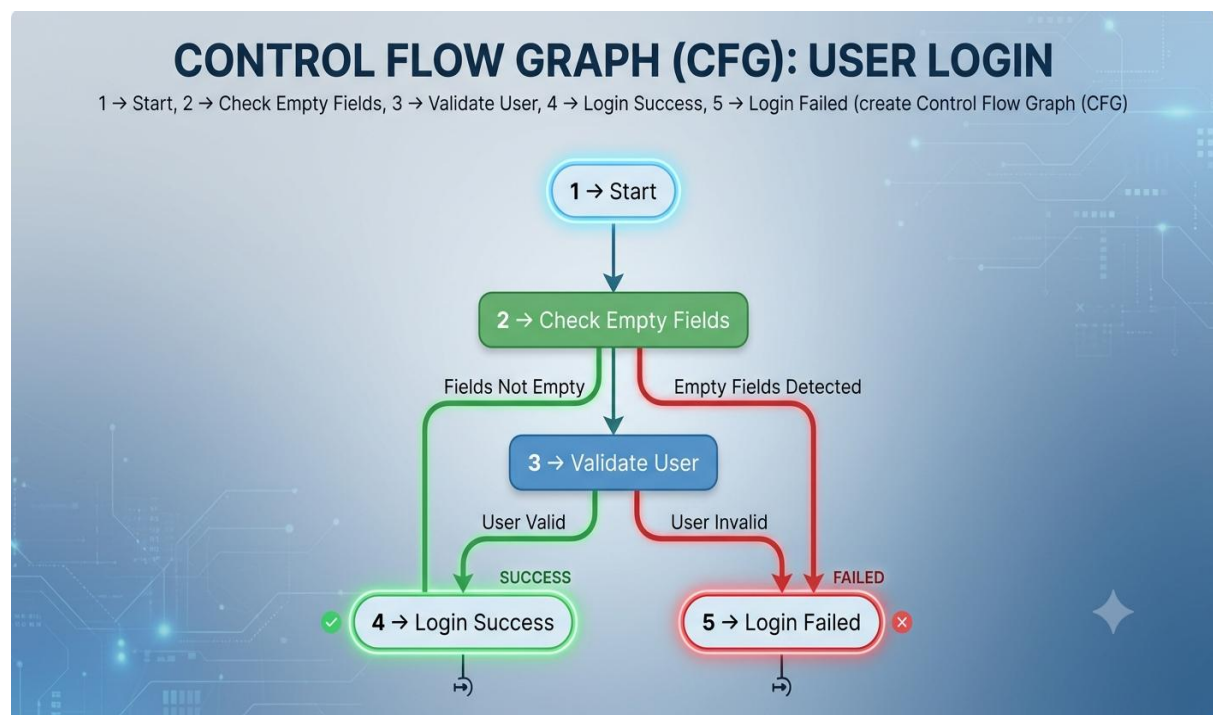
13.1 Login Module

Pseudocode

```
</> Java

if(username.isEmpty() || password.isEmpty())
    displayError();
else if(validUser())
    loginSuccess();
else
    loginFailed();
```

Control Flow Graph (CFG)



Cyclomatic Complexity

Formula:

$$V(G) = E - N + 2$$

Or

$$V(G) = \text{Number of Decision Nodes} + 1$$

Decision Nodes = 2

Therefore,

$$V(G) = 2 + 1$$

$$V(G) = 3$$

Independent Paths

- Path 1

Start → Empty Fields → Error Message

- Path 2

Start → Valid User → Login Success

- Path 3

Start → Invalid User → Login Failed

Statement Coverage

$$\text{Coverage} = (\text{Executed Statements} / \text{Total Statements}) \times 100$$

Coverage Achieved:

100%

Decision Coverage

$$\text{Coverage} = (\text{Executed Decisions} / \text{Total Decisions}) \times 100$$

Coverage Achieved:

100%

14. Defect Report

Defect ID	Description	Severity	Status
D01	Login accepts empty fields	Medium	Fixed
D02	Invoice formatting issue	Low	Fixed
D03	Duplicate customer records	Medium	Fixed

15. Test Summary Report

Item	Result
Total Test Cases	6
Passed	6
Failed	0
Pass Percentage	100%

16. Conclusion

The Green Leaf Bakery System was successfully tested using both Black Box and White Box testing techniques. All functional requirements were verified, and the system performed according to the expected results.

The White Box Testing techniques confirmed that the major modules achieved full statement and decision coverage. Therefore, the Green Leaf Bakery System is considered reliable, functional, and ready for deployment.

17. References

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